

Required Assignment 5.1: Will the Customer Accept the Coupon?

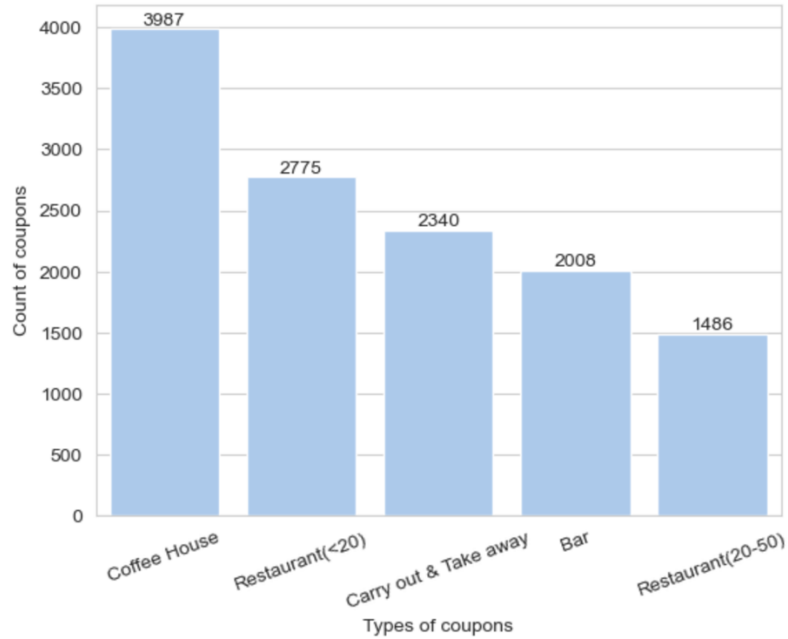
Plot Summary & Plot attached (Page 2)

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1. Coupons for cheaper restaurants (e.g., Restaurant <20) have a significantly higher acceptance rate compared to bar coupons, indicating price sensitivity among customers.
2. Coupons distributed at 80°F are more likely to be accepted, suggesting that weather conditions, possibly associated with leisure activities, influence acceptance.
3. Coupons distributed at 2 PM have the highest probability of acceptance, aligning with typical meal or break times, while early morning and late-night distribution times yield lower acceptance rates.
4. Coupons with a 1-day expiration period have a higher likelihood of acceptance than those with shorter expiration windows, suggesting that a reasonable time to redeem adds value for customers.
5. Drivers traveling in the opposite direction of the coupon location are more inclined to accept, potentially driven by flexibility in route or intent to visit the location.
6. Single and married partners show higher acceptance rates, suggesting a potential correlation with relationship status and leisure or dining behavior.
7. Drivers visiting bars three or fewer times are more likely to accept bar coupons, suggesting infrequent visitors are more incentivized by discounts.
8. Drivers over 25, without kid passengers, and in non-farming occupations, have the highest bar coupon acceptance rates, indicating that demographics and context heavily influence bar coupon success.
9. Female drivers with kid passengers have a higher acceptance rate for coupons compared to male drivers in similar scenarios, indicating gendered behavior towards convenience or planning.
10. Younger drivers (21, 26, 31) and older drivers (50+) are offered more coupons than middle-aged drivers (36–46), possibly due to targeting strategies based on perceived activity levels and consumer behavior.
11. Drivers are more likely to accept coupons from cheap restaurants than bars, demonstrating a preference for economical dining options.

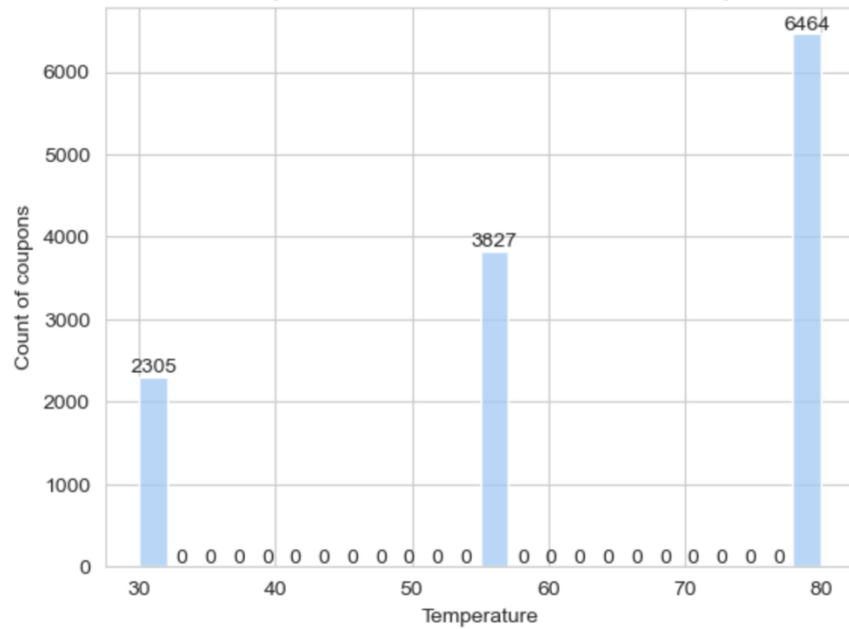
Coupon distribution

Distribution by types of coupon

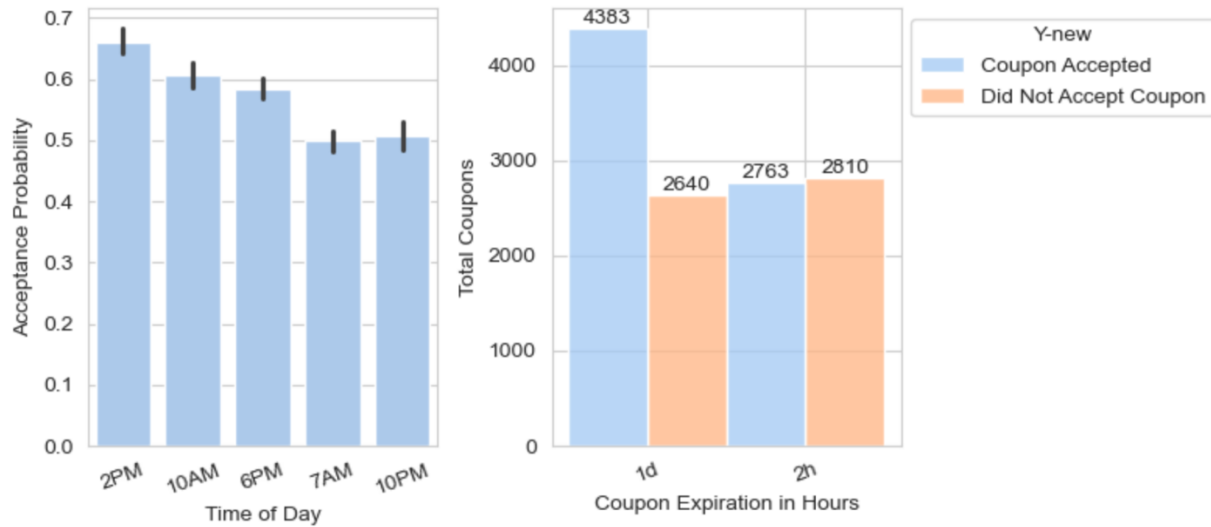


Coupon distribution by temperature

Count of coupons distributed at various recorded temperatures

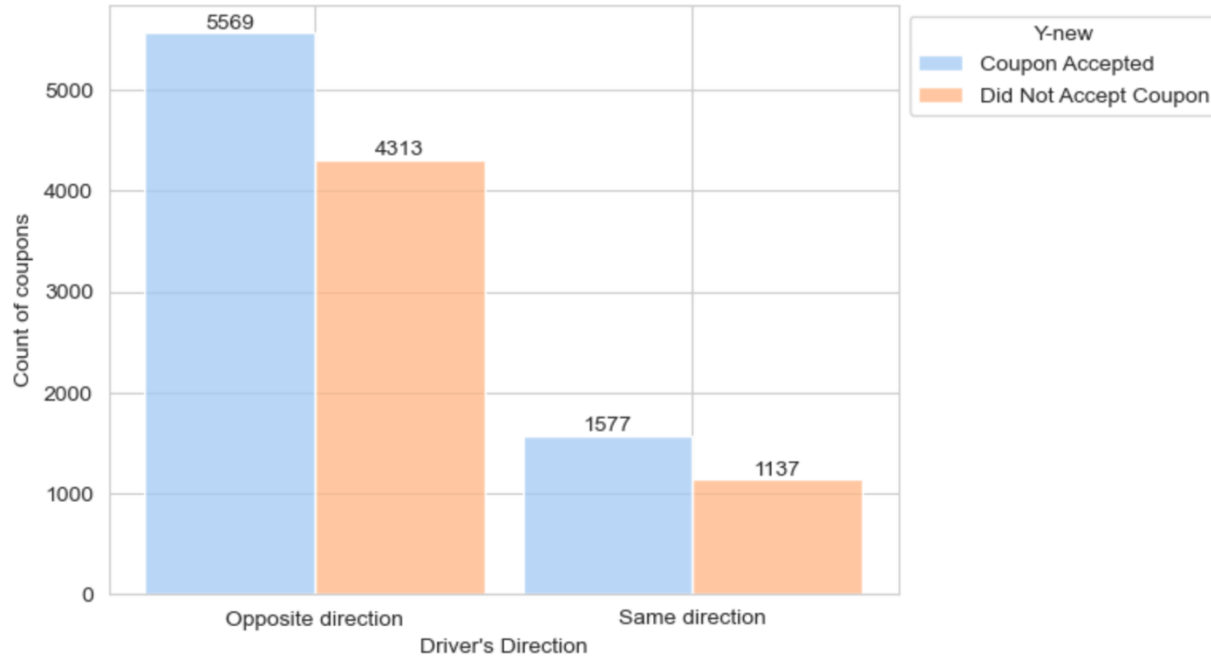


Time of Day, Coupon Expiration - Acceptance Ratio

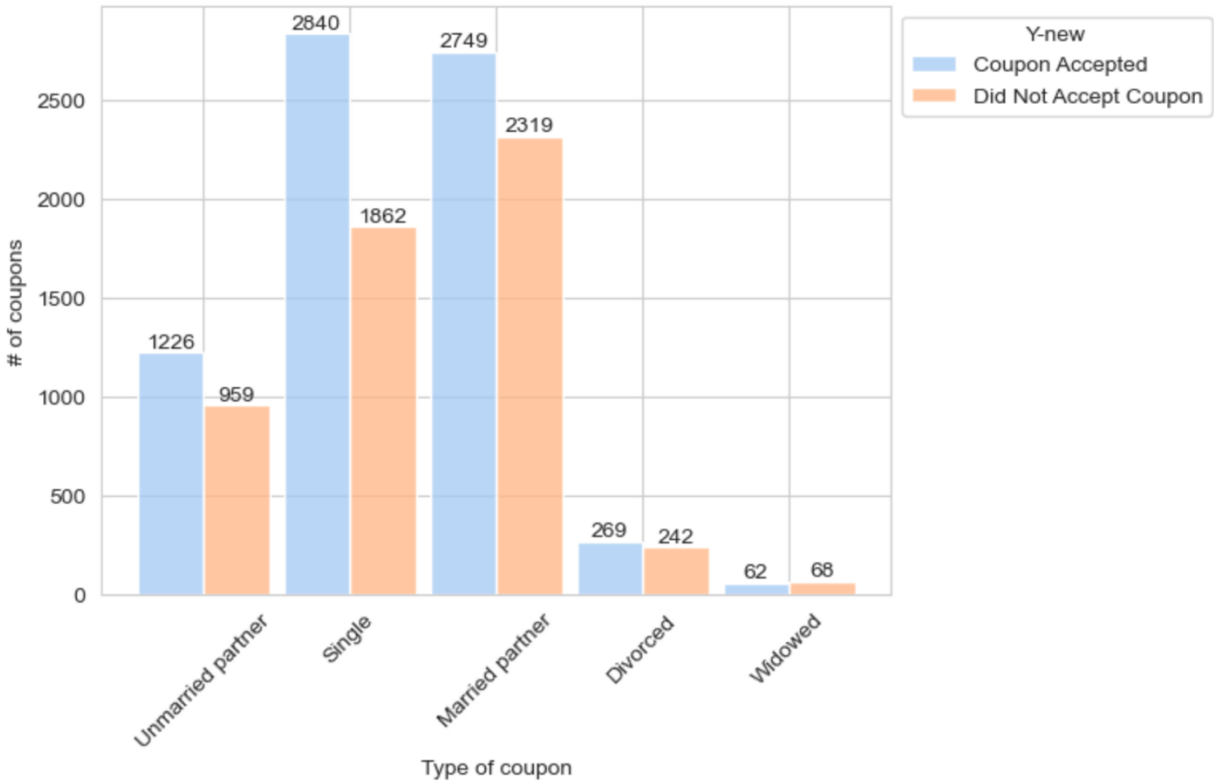


Coupon acceptance by direction

Drivers in opposite direction were offered and accepted more coupons

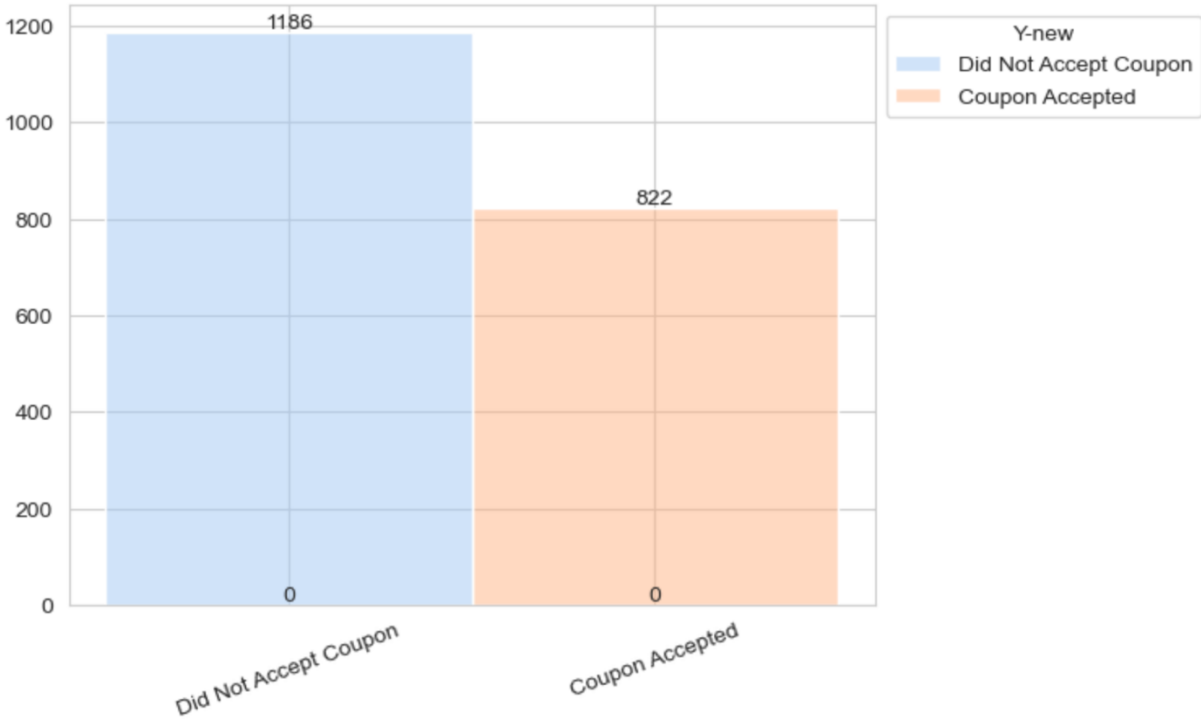


Coupons accepted by Martial Status



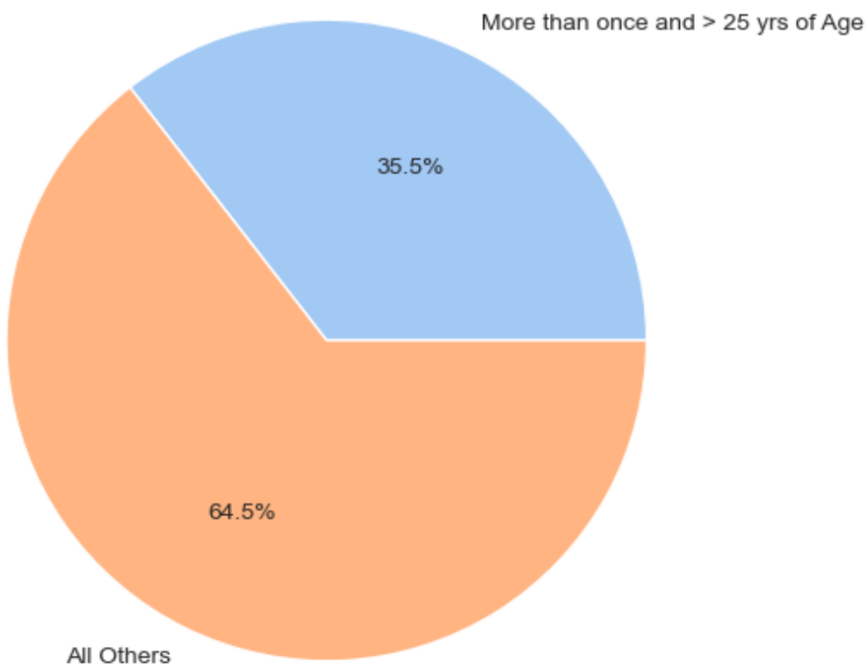
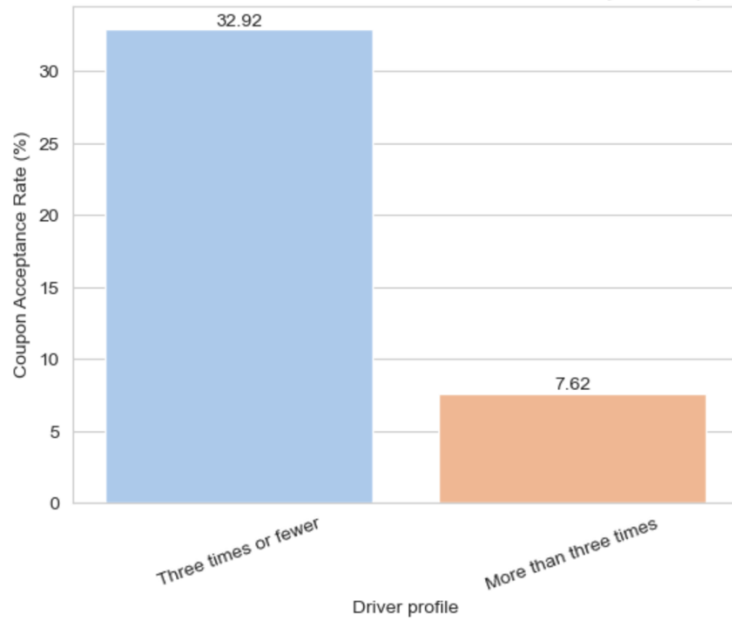
Bar coupons

Proportion of bar coupons accepted: 40.94%



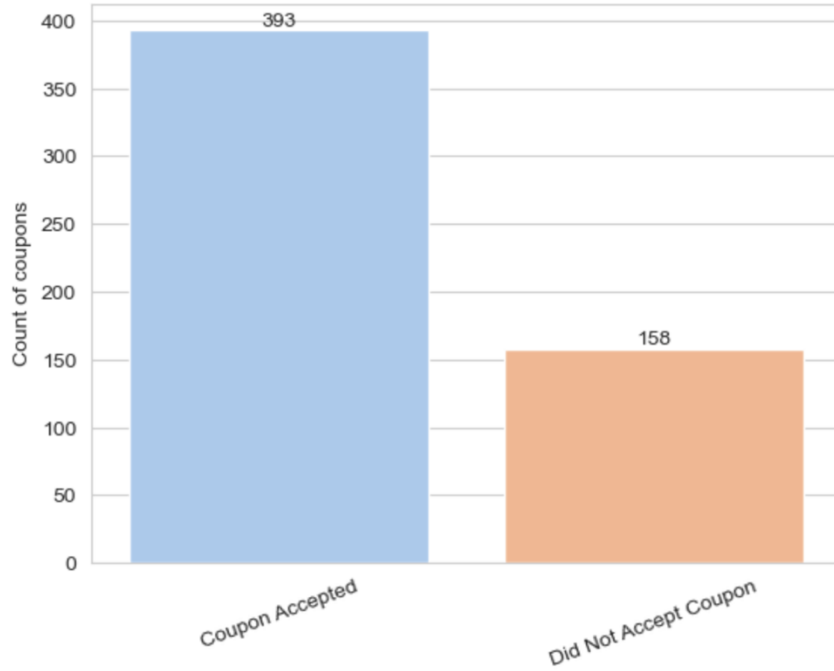
Coupon acceptance rates amongst Bar visitors

Drivers who went to bar three or fewer times were 4.32 more likely to accept a bar coupon



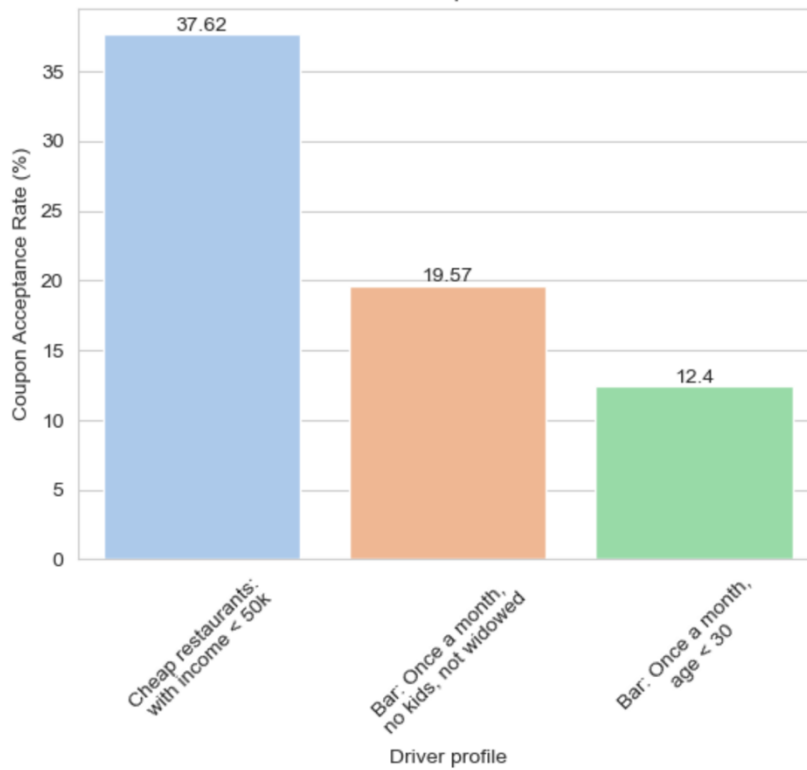
Acceptance rates of drivers to Bars

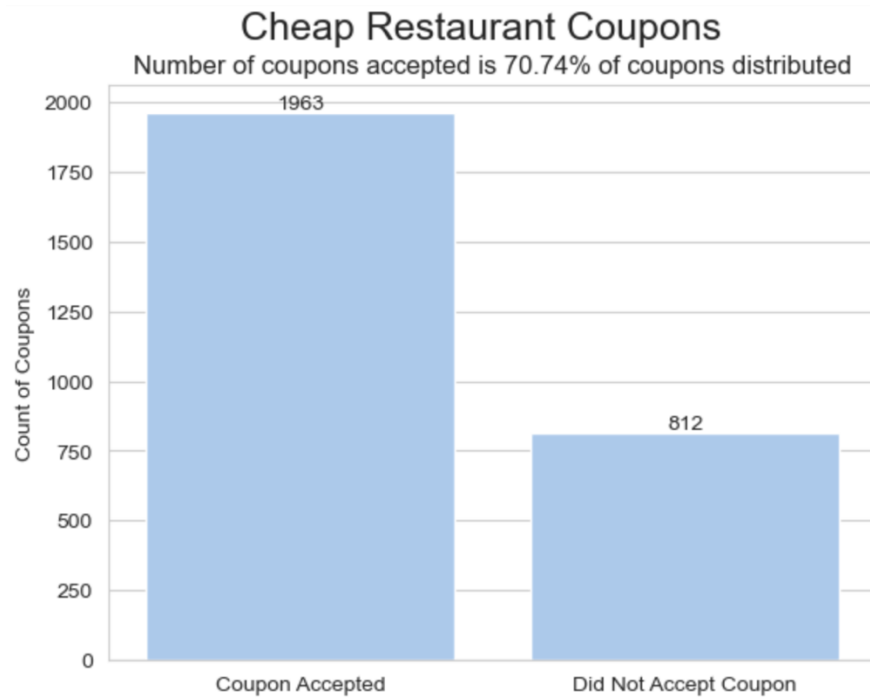
more than once a month, no kids passengers, non farming occupations



Coupon acceptance rates of drivers

Bars and Cheap Restaurants





Cheap Restaurant Coupons

Distribution and acceptance of coupons by age

